OGOMA'S LEMMA: REALIZING AN ANNIHILATOR EQUALITY OVER THE LOCALIZATION BY MULTIPLYING WITH AN ELEMENT OF S

THE STACKS PROJECT AUTHORS

Editorial note. The source proof is complete; no omitted steps.

Lemma 0.1 (Ogoma)**.** *Let A be a Noetherian ring and let M be a finite A-module. Let $S \subset A$ be a multiplicative set. If $\pi \in A$ and $\mathrm{Ker}(\pi : S^{-1}M \to S^{-1}M) = \mathrm{Ker}(\pi^2 : S^{-1}M \to S^{-1}M)$ then there exists an $s \in S$ such that for any $n > 0$ we have $\mathrm{Ker}(s^n\pi : M \to M) = \mathrm{Ker}((s^n\pi)^2 : M \to M)$.*

Proof. Let $K = \mathrm{Ker}(\pi : M \to M)$ and $K' = \{m \in M \mid \pi^2 m = 0 \text{ in } S^{-1}M\}$ and $Q = K'/K$. Note that $S^{-1}Q = 0$ by assumption. Since A is Noetherian we see that Q is a finite A-module. Hence we can find an $s \in S$ such that s annihilates Q. Then s works. □

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